In total we used the 2,544 recent shallow landslide initiation observations (i.e., pixels) (Figure 3) and an equal number of random points taken across the landscape outside the landslide areas that served as 'non-landslide' locations. The model was trained twice; once with the morphological data (elevation, slope, curvature, exposure) derived from the 30 m SRTM elevation data, and one time with the same morphological data derived from the 10 m resolution DEM for Rwanda. We apply a forward stepwise regression in order to obtain the highest model quality using the lowest possible number of predictors (Hosmer and Lemeshow, 2000). We used the area under the ROC (AUC) curve as a metric for model quality (e.g., Jacobs et al., 2018; Depicker et al., 2020b). The AUC is a measure of the discriminatory power of the model, i.e. the probability that a random landslide location will receive a higher susceptibility score than a random non-landslide location.

Following the approach proposed by Depicker et al. (2020b), we applied 10-fold cross-validation in order to ensure that our model is not overfit (i.e. it works well for the training data of the susceptibility model, but performs poorly on locations that were not used for model training). Hence, we randomly split our data in 10 groups of equal size. Subsequently, we tried to predict the susceptibility for each group and assess the associated AUC value, each time using the other nine groups for model training. We reported the average AUC value of these 10 observations. The validation shows that overall the model performance are better with the 30m SRTM (Table 2). Data-driven models are the most performant when a good balance between data resolution and model complexity is kept. This is something that we have already observed for the landslide susceptibility assessment in the Rwenzori mountains (Jacobs et al., 2018).

Table 2: **Validation of the susceptibility models using a 10-fold cross validation procedure**

Model	AUC (10-fold CV) %	
	lithology	No lithology
10 m DEM	89.3	85.0
30 m DEM	91.1	88.8

The difference of model prediction performance is marginal whether the lithology is used or not. As the lithological information is coarse at the watershed scale, we selected the model without lithology (highlighted in yellow in Table 2).

To summarize, we used in this study the regional susceptibility model that was computed with all the predictor variables but the lithology at a 30m resolution (Table 2). The hazard assessment (Step 3) is based on this susceptibility model.